HACKING THE SMART OFFICE: EXPLOITING VoIP AND IoT CAMERAS

THEY SEE.
THEY LISTEN.
THEY OWN.

UNAUTHORIZED ACCESS

CAMERA_01	ONLINE
VOIP_PHONE	COMPROMISED
NETWORK	OWNED

MODERN OFFICE. HIDDEN WEAKNESSES. REAL CONSEQUENCES.
ATTACKERS DON'T NEED PHYSICAL ACCESS TO TAKE OVER YOUR OFFICE.

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EVERY DEVICE IS AN ENTRY POINT
VoIP phones and IoT cameras are often misconfigured and poorly monitored.

BUILT FOR CONVENIENCE, ABUSED FOR ACCESS
Designed for ease of use, not for attackers with bad intentions.

SILENT. STEALTHY. DEVASTATING.
Eavesdrop on calls. Watch everything. Move laterally. Steal it all.

YOUR OFFICE IS THEIR PLAYGROUND
From lobby cams to boardroom calls—attackers stay hidden and in control.

IT'S NOT IF. IT'S WHEN.
Secure today, or be the next headline tomorrow.

HOW ATTACKERS TAKE OVER (KILL CHAIN)

1. RECONNAISSANCE → 2. ACCESS → 3. PERSISTENCE → 4. SURVEILLANCE → 5. EXFILTRATION

1. **RECONNAISSANCE** – Scan for open VoIP ports, SIP services, and exposed cameras.
2. **ACCESS** – Exploit default creds, weak auth, or known vulnerabilities.
3. **PERSISTENCE** – Create backdoors, add rogue accounts, maintain access.
4. **SURVEILLANCE** – Access camera feeds, eavesdrop on calls, monitor activity.
5. **EXFILTRATION** – Steal data, recordings, credentials, and sensitive documents.

THEY DON'T BREAK IN. THEY LOG IN.

ATTACK VECTORS

VoIP SYSTEMS
- SIP enumeration & brute force
- Default/Fabricated credentials
- SIP INVITE/REGISTER spoofing
- Voicemail PIN brute force
- Call recording interception
- VoIP device firmware exploits

IoT CAMERAS
- Default credentials
- ONVIF service abuse
- Unencrypted RTSP streams
- Firmware vulnerabilities
- Exposed cloud services
- Motion events as side-channel

ONE COMPROMISED DEVICE CAN LEAD TO EVERYTHING.

REAL-WORLD IMPACT

ESPIONAGE
Watch meetings, track employees, steal trade secrets.

FINANCIAL LOSS
Fraud, wire transfer interception, ransomware, data theft.

REPUTATIONAL DAMAGE
Loss of trust, legal consequences, regulatory fines.

PHYSICAL SAFETY RISK
Stalking, workplace threats, unwanted surveillance.

REAL-WORLD EXAMPLES

Ring & Hikvision Breaches
Exposed cameras were used to spy on homes and offices worldwide.

VoIP Spam & Toll Fraud
Attackers abused PBX systems to make expensive international calls.

SIP Credential Leaks
Leaked VoIP credentials sold on dark web for lateral movement.

IoT Botnets
Compromised cameras used in DDoS attacks (e.g., Mirai, Mozi).

WHAT ATTACKERS CAN DO

- Live view & record all cameras
- Eavesdrop & record phone calls
- Extract voicemails & call logs
- Move laterally in internal network
- Steal documents, credentials, data
- Deploy ransomware or wipers

AT RISK

- VoIP Phones & PBX
- IP Cameras & NVRs
- SIP Gateways & ATA
- Conference Systems
- Smart Displays
- Access Control IoT
- Environmental Sensors

DEFEND YOUR SMART OFFICE

CHANGE DEFAULT CREDENTIALS
Enforce strong, unique passwords on all devices.

NETWORK SEGMENTATION
Isolate VoIP and IoT devices in separate VLANs.

ENCRYPT COMMUNICATIONS
Use SRTP for VoIP and HTTPS/SSL for device access.

KEEP FIRMWARE UPDATED
Regularly update firmware and patch known vulnerabilities.

MONITOR & DETECT
Monitor SIP traffic, device behavior, and anomalies.

ACCESS CONTROL
Restrict access to management interfaces.

DISABLE UNNEEDED SERVICES
Turn off UPnP, Telnet, ONVIF (if not needed), and cloud.

TOOLS ATTACKERS USE

- SIPVicious / sipsak – SIP enumeration & fuzzing
- INVITEFLOOD – SIP brute force & DoS
- Linphone / Zoiper – VoIP call interception
- Nmap / Masscan – Port scanning
- ONVIF Device Manager – Camera discovery & abuse
- ffmpeg / VLC – Stream grabbing (RTSP)
- Hydra / Medusa – Credential brute forcing
- Burp Suite / Wireshark – Traffic analysis

QUICK SECURITY CHECKLIST

- ☑ Change all default passwords
- ☑ Enable strong authentication
- ☑ Encrypt VoIP (SRTP) & Web access (HTTPS)
- ☑ Segment VoIP/IoT from corporate network
- ☑ Disable unused services & ports
- ☑ Enable logging & fail2ban
- ☑ Backup configs & monitoring in place
- ☑ Regularly audit devices & logs

THE BOTTOM LINE

IN THE SMART OFFICE, CONVENIENCE BUILDS THE ATTACK SURFACE.

SECURE TODAY. OR REGRET TOMORROW.

THEY DON'T NEED TO BREAK DOWN THE DOOR.
THEY JUST NEED A WEAK DEVICE INSIDE.